

MUHAMMAD ASAD

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PROFESSIONAL SUMMARY

Second-year Computer Software Engineering student at Bahria University and Backend Development Intern at ByteCorp Technologies, building a production-style Django REST Framework job board backend with JWT authentication and role-based access control. Experienced across Python/Django, Java, C#.NET, and JavaScript, with 15+ open-source projects spanning REST APIs, AI-powered tools (ResumeAI), and full-stack applications. Comfortable working with Git-based, PR-driven workflows and mentor code review.

EDUCATION

Bachelor of Engineering in Computer Software Engineering - Sep 2024 - Sep 2028

Bahria University - Karachi Campus

Intermediate - Computer Science (FSc) - Completed

Malir Cantt College

EXPERIENCE

Backend Development Trainee (Remote) - June 2026 - Present (3 Months)

ByteCorp Technologies - Training Program

- Collaborating with a technical mentor (Midha Tahir) on the end-to-end development of a production-style Django REST Framework job board backend, from database design through API implementation and security.
- Delivered Phase 1 on schedule: PostgreSQL schema design and ERD, five-app Django architecture (accounts, companies, jobs, job applications, skills), REST API implementation with JWT authentication (15-min access / 7-day refresh tokens), role-based access control across 3 roles (job seeker, company rep, admin), and ownership checks via a CompanyMember model.
- Currently implementing strict validation (Section 8) across all five modules with DRF serializer- and business-rule-level checks, following a structured PR-based workflow with mentor review.
- Sat and passed a technical viva covering API design (REST), authentication (JWT), and authorization (RBAC + ownership).

Java Development Intern (Remote) - 2026

AICTE Oasis Infobyte Internship Program (OIBSIP) - Java Development Track

- Completing a 5-task Java internship track; built an ATM Interface application using Java OOP and ArrayList-based transaction handling, and an Online Examination System with a custom Java Swing UI and timer-based test flow.
- Following structured, task-wise submission guidelines with documented code and demo walkthroughs for each task, pushed to a dedicated GitHub repository.

Software Development Intern (Remote) - 2024

CodeAlpha - Project-Based Internship

- Delivered two production-quality Java applications (Student Grade Manager, Hotel Reservation System) under real internship specifications; both reviewed on GitHub.
- Applied clean code principles, OOP design patterns, Git version control, and project documentation throughout a 4-week engagement.
- Both submissions adopted as reference implementations on the internship platform following positive architecture review.

PROJECTS

ResumeAI | FastAPI, Next.js 15, TypeScript, PostgreSQL, SQLAlchemy, JWT, OpenAI/OpenRouter API, Docker | GitHub

- Built a production-ready AI-powered resume analysis platform with ATS scoring, skill-gap analysis, job-description matching, and AI-driven improvement suggestions.
- Designed a FastAPI backend with 18+ REST API endpoints across authentication, resume, analysis, and job-matching modules, JWT access/refresh authentication, Alembic-managed PostgreSQL schema, rate limiting, and Pydantic validation, paired with a Next.js/TypeScript frontend and an analytics dashboard.
- Integrated LLM-based analysis via OpenRouter with centralized retry logic and exponential backoff for transient errors, and containerized the full stack with Docker Compose for one-command deployment.

FoodHeaven - Restaurant POS System | Python, Django REST Framework, PostgreSQL, JavaScript, PyWebView | GitHub

- Built a native desktop Point of Sale system for restaurants with a dark-themed register UI, real-time cart management, GST calculation, and a Kitchen Monitor Display for live order tracking.
- Developed an Analytics Dashboard using Chart.js for sales trends, along with printable invoice generation and multi-mode checkout backed by a PostgreSQL database across modular Django apps.

Design Patterns in Java & Python | Java, Python, OOP, Software Design | GitHub

- Studied and implemented all 23 GoF design patterns (creational, structural, behavioral) with working code examples in both Java and Python, pushed to a dedicated open-source repository.

Food Heaven Website | HTML, CSS, JavaScript, Bootstrap 5, .NET Framework | GitHub

- Built a premium restaurant management system featuring a modern UI, dynamic menu display, and a robust table reservation system with responsive design.
- Implemented client-side interactivity using vanilla JavaScript and Bootstrap 5 grid for a seamless user experience across all devices.

Algo Puzzle Board | HTML, CSS, JavaScript, Data Structures & Algorithms, .NET Framework | GitHub

- Engineered an interactive algorithm visualization platform covering 9 classic algorithms: sorting, pathfinding, N-Queens, Knight's Tour, Graph Coloring, TSP, Huffman Coding, Dijkstra's, and Kruskal's MST.
- Designed modular dashboard architecture with animated step-by-step algorithm walkthroughs to aid conceptual understanding.

Requirement Genie | Python, NLP, AI/ML, Streamlit | GitHub

- Developed an AI-driven requirements engineering assistant that automates the generation of SRS documents using NLP techniques.

- Integrated a conversational interface for systematic requirements gathering, reducing manual documentation effort significantly.

Job Ranking System (SkillMesh) | Python, DSA, Greedy Algorithms, OOP | GitHub

- Built a DSA-based smart candidate ranking and recommendation engine that scores job applicants using skill matching, experience scoring, and greedy budget optimization algorithms.
- Implemented a Max Heap data structure for efficient top-candidate retrieval with an Explain Mode for step-by-step algorithm tracing.

Decode & Discover | JavaScript, HTML, CSS, Cryptography | GitHub

- Developed an educational cryptography tool that teaches core security concepts through interactive encryption and decryption visualizations with a narrative-driven story format.
- Designed three distinct story-based learning modules to engage users while reinforcing cipher and encoding principles.

MIPS Productivity Dashboard | MIPS Assembly, Computer Architecture | GitHub

- Built a low-level task management system entirely in MIPS Assembly language featuring task tracking, calendar view, and direct memory status updates.
- Demonstrated deep understanding of computer architecture, register management, and memory-mapped I/O in a resource-constrained environment.

Snake Game | Python, OOP, Pygame / JavaScript | GitHub

- Implemented a classic retro-style Snake Game with smooth controls, real-time score tracking, collision detection logic, and progressive difficulty scaling.

Python Streamlit Dashboard Suite | Python, Streamlit, Pandas, Matplotlib | GitHub

- Developed 3+ interactive web dashboards with real-time data filtering, dynamic charting, and multi-source dataset aggregation using Pandas and Streamlit widgets.
- Self-taught Streamlit beyond curriculum; published deployable dashboard apps demonstrating data wrangling and visualization skills.

Student Grade Tracker & Manager | Java, OOP, Collections Framework | GitHub

- Built a Java console application with CRUD operations, automated grade computation (average, highest, lowest), and formatted report generation.
- Implemented encapsulation, inheritance, and polymorphism for a clean, extensible OOP architecture; delivered as a CodeAlpha internship project.

Hotel Reservation System | Java, OOP, ArrayList, Input Validation | GitHub

- Designed a hotel management application supporting room booking, guest registration, real-time availability checks, and full reservation lifecycle management.
- Modeled domain entities (Room, Guest, Booking) as Java classes with robust input validation and error handling for production-grade reliability.

Python Mini-Projects Collection (10+) | Python, OOP, File I/O | GitHub

- Delivered 10+ Python applications including To-Do Manager, Calculator, Quiz Engine, and Number Guesser, each reinforcing OOP design, file handling, and clean code practices.

TECHNICAL SKILLS

Languages: C#, Java, Python, JavaScript, MIPS Assembly, SQL, HTML, CSS

Frameworks & Libraries: Bootstrap 5, Streamlit, Pandas, Matplotlib, NumPy, Java Swing, C# Windows Forms, Django, Django REST Framework, PyWebView, Chart.js

Databases: MySQL, SQL Server, SQLite, PostgreSQL, Oracle

APIs & Authentication: REST APIs, FastAPI, Django REST Framework, JWT Authentication, Role-Based Access Control (RBAC), Ownership-Based Authorization

Developer Tools: Git, GitHub, VS Code, Visual Studio, IntelliJ IDEA, PyCharm, Netlify

Core Concepts: OOP, Data Structures & Algorithms, Design and Analysis of Algorithms, REST APIs, NLP, Computer Architecture, Agile Basics

Software Architecture & Design: Design Patterns (23 GoF Patterns), Design Principles (SOLID), MVC, Client-Server, Layered, Peer-to-Peer, Pipe-and-Filter, etc.

ACHIEVEMENTS & HIGHLIGHTS

- 15+ open-source projects on GitHub across Java, Python, C#, and JavaScript with a consistent commit history demonstrating real-world development discipline.
- Currently completing a Backend Development Internship at ByteCorp Technologies (ongoing), building a production-style Django REST Framework job board backend.
- Currently completing a Java Development Internship under the AICTE OIBSIP program (ongoing), building an ATM Interface and an Online Examination System.
- Implemented all 23 GoF design patterns in both Java and Python as a dedicated open-source reference repository.
- Built ResumeAI, a full-stack AI resume analysis platform (FastAPI + Next.js) with LLM-powered ATS scoring and job-matching, deployed via Docker.
- Portfolio deployed on Netlify showcasing 10+ production-quality projects spanning web development, AI tools, algorithm visualizations, and systems programming.
- CodeAlpha Internship: completed a remote software development internship, delivering enterprise-style Java applications under real project deadlines.
- Self-directed learning: independently mastered Streamlit, Pandas, NLP basics, and data visualization tools beyond coursework; shipped multiple deployable applications.
- Built an AI-powered requirements engineering tool (Requirement Genie) demonstrating initiative in applying NLP to real software engineering problems.